

C# BACKEND TRACK | OOP PHASE

Task 1

System Analysis Task

Think about the business. Extract the models. Assign the sources.

Codeline Bootcamp · Backend Phase · OOP Session 3 Task

SOURCE TAG LEGEND

Every property of every class model gets its value from somewhere. Before writing any code, you must decide **where** each value comes from. Use one of these five source tags to label each property in your analysis:

System Generated	System Generated	The system assigns this value automatically. The user never types it. Example: an auto-incremented ID calculated as List.Count + 1.
User Input	User Input	The user types this value directly at the keyboard when prompted by the system.
Default Value	Default Value	A fixed value set by the code when the object is created. The user never chooses it. Example: isBooked = false, status = "Active".
Calculated	Calculated	Derived automatically from another object already in memory — no user prompt. Example: appointment date copied from a chosen slot, fee copied from a doctor.
From List	From List	The user selects a value from a list of options displayed on screen. Example: choosing a doctor ID from a printed list of doctors.

Domain 1 — Library Management System

Extract everything yourself

Read the following description of a real library carefully. Your job is to go through the 6-phase process and fill in all the blank tables below. Nothing has been done for you — you identify the entities, you decide the properties, and you assign the source tag for each one.

A public library allows community members to register and borrow books. Each book has a title, author, genre, and a fixed number of physical copies in stock. A registered member can borrow an available book copy and must return it within 14 days. If the book is returned late, the system calculates a fine based on how many days overdue it is. A member can also reserve a book that is currently fully borrowed so they are notified when it becomes available. The librarian needs to see all currently active loans, view all overdue loans sorted by how late they are, and get a report of the most-borrowed books ranked by total borrow count.

Step 1 — Identify the Entities (Phases 1 & 2)

From the description above, identify the main real-world objects (classes) the system must track. Write each entity name, its properties, the data type for each, and the source tag that describes how each property gets its value.

Entity Name: <u>Books</u>		
Property Name	Data Type	Source Tag
<u>BookID</u>	<u>int</u>	<u>system generated</u>
<u>Auther</u>	<u>string</u>	<u>user input</u>
<u>Genre</u>	<u>string</u>	<u>user input</u>
<u>Titel</u>	<u>string</u>	<u>user input</u>
<u>TotalCopies</u>	<u>int</u>	<u>Calculate</u>
<u>availble copies</u>	<u>int</u>	<u>calculated</u>
<u></u>	<u></u>	<u></u>

Entity Name: <u>RegisteredMember</u>		
Property Name	Data Type	Source Tag
<u>MemberID</u>	<u>int</u>	<u>system generated</u>
<u>Name</u>	<u>string</u>	<u>user input</u>
<u>Phone</u>	<u>int</u>	<u>user input</u>
<u>email</u>	<u>string</u>	<u>user input</u>
<u>registerdDate</u>	<u>string</u>	<u>system generated</u>
<u>Activation</u>	<u>bool</u>	<u>default value</u>
<u></u>	<u></u>	<u></u>

Entity Name: <u>loans</u>		
Property Name	Data Type	Source Tag
<u>LoneID</u>	<u>int</u>	<u>system gnreated</u>
<u>MemberID</u>	<u>int</u>	<u>from list</u>
<u>bookID</u>	<u>int</u>	<u>from list</u>
<u>borrowDate</u>	<u>string</u>	<u>system generated</u>
<u>dueDate</u>	<u>string</u>	<u>system generated</u>
<u>returnDate</u>	<u>string</u>	<u>system generated</u>
<u>isReturned</u>	<u>bool</u>	<u>default value</u>

Entity Name: <u>Librarian</u>		
Property Name	Data Type	Source Tag
<u>LibrarianID</u>	<u>int</u>	<u>System generated</u>
<u>activeLone</u>	<u>int</u>	<u>calculated</u>
<u>overDueLoans</u>	<u>int</u>	<u>calculated</u>
<u>mostBorrowBooks</u>	<u>string</u>	<u>calculated</u>
<u>totalBorrowCount</u>	<u>string</u>	<u>default value</u>
<u></u>	<u></u>	<u></u>
<u></u>	<u></u>	<u></u>

If you identify more than 4 entities, continue on the back of this sheet or add extra rows above.

Step 2 — Describe the Entity Relationships (Phase 1 continued)

How do the entities connect to each other? Describe the full cycle in plain English — who does what, which entity gets created, and what changes in other entities as a result.

The member enters his name, the system gives him a memberId automatically, the librarian checks if the member is Active. After that the member enters his details and enters the book title, the system checks if the book is available if it is available the member takes a Loan and the system records the borrow date, calculates the due date automatically and decreases one copy from the stock. When the member returns the book, the librarian checks if he returned it on time if on time the system closes the Loan and adds the copy back to the stock, if late the system calculates a fine automatically based on the number of overdue days.

Step 3 — List the Services (Phase 4)

From the description, identify every service (function) the system needs. Write at least 8 services — give each a name and a one-line description of what it does.

#	Service Name	What it does
1	Register member	When member registered the system give him id automatically
2	Borrow book	Add loan for member if book still not returned
3	returnBook	The loan cancelled if the book returned
4	ReciveBook	Creates a Reservation for a member if the book is fully borrowed
5	addBooks	the system add book automatically in books list
6		
7		
8		
9		
10		

Domain 2 — Hotel Management System

Assign the source tags

The class models and all their properties are already defined for you below. Your job is to **fill in the Source Tag column** for every single property. Use the five source tags from the legend on page 1: System Generated, User Input, Default Value, Calculated, From List.

A hotel allows guests to register and book rooms for a stay. The hotel has different room types (Single, Double, Suite) each with its own nightly price. Each room can only have one active booking at a time. When a guest checks in, a check-in record is created. The total price of a stay is calculated from the number of nights and the room's nightly rate. When a guest checks out, the room becomes available again. A booking can also be cancelled before check-in, which frees the room immediately.

Your task: For each property in the tables below, write the correct source tag in the empty column on the right. Think carefully — ask yourself: does the user type this? does the system generate it? is it copied from another object? does it have a fixed starting value? does the user pick it from a list?

Guest		
Property Name	Data Type	Source Tag (fill in)
<code>guestId</code>	int	System generated
<code>guestName</code>	string	User input
<code>guestPhone</code>	string	User input
<code>guestEmail</code>	string	User input
<code>guestNationality</code>	string	User input
<code>registrationDate</code>	string	Calculated

Room		
Property Name	Data Type	Source Tag (fill in)
roomId	int	System generated
roomNumber	string	user input
roomType	string	From list
floorNumber	int	From list
nightlyPrice	decimal	Calculated
isAvailable	bool	Default value

Booking		
Property Name	Data Type	Source Tag (fill in)
bookingId	int	System generated
guestId	int	user input
roomId	int	User input
checkInDate	string	User input
checkOutDate	string	User input
numberOfNights	int	From list
totalPrice	decimal	Calculated
status	string	Default value

CheckInRecord		
Property Name	Data Type	Source Tag (fill in)
recordId	int	System generated
bookingId	int	User input
guestId	int	User input
roomId	int	User input
actualCheckIn	string	User input
actualCheckOut	string	User input
finalAmount	decimal	Calculated

Step 2 — Describe the Entity Relationships

Now that you have studied the four class models, describe in your own words how they connect to each other. Who creates what? What changes when a booking is made? What happens when a guest checks out?

Write the relationship cycle here:

Guest enters his name and the system generate automatically id for him then choice the room type and total night after that system check if the room type is available.
after that the system gives him room and calculate the total price then he conform booking . however, the system they give him record id to check In with check-in date&Tima and checkout date&Time.

Step 3 — List the Services

Based on the description and the class models above, list all the services (functions) this hotel system needs. Give each a name and a one-line description.

#	Service Name	What it does
1	Guest name	The guest must enter his name and system give him ID
2	RoomType	They choice room type
3	isAvailable	system check if the room available or not
4	priceing	After choice room type and total night system calculate price
5	Booking	he will get conformation for booking with booking id
6	Recording	They will give him check-in date and time with check-out time and date
7		

#	Service Name	What it does
8		
9		